SMALL MAXIMAL k-CONSECUTIVE SUMS ON A CIRCLE AND THE CLOCK-FACE PROBLEM

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ABSTRACT. Arrange $1, 2, \ldots, n$ on a circle and let $\mathsf{msum}(n, k)$ be the smallest possible value of the largest sum of k consecutive entries, minus the average $k(n+1)/2$ of these sums. For odd $k \geq 3$ and odd m, Higashitani and Kurimoto proved $\mathsf{msum}(mk, k) = 2$ for $m \geq 2k+3$, while for $5 \leq m \leq 2k+1$ it was only known that $\mathsf{msum}(mk, k) \in \{1, 2\}$, apart from two cases with $k = 3$. We show that $\mathsf{msum}(mk, k) = 1$ for $3 \leq m \leq k+2$ and $\mathsf{msum}(mk, k) = 2$ for $m \geq k+4$, which settles this case. The lower bound comes from a short total-variation argument giving, for odd k, $\mathsf{msum}(n, k) \geq 2$ for every odd $n > (k+3)\gcd(n, k)$; the upper bound is an explicit construction. For $k = 3$ the argument gives a short proof, with no computer verification, that the clock-face numbers (OEIS A066385) are at least $\lceil 3n/2 \rceil + 3$ for all $n \geq 4$ except $n = 5, 6, 9, 15$. With the known matching upper bound this gives $a(n) = \lceil 3n/2 \rceil + 3$ apart from these four exceptions, as conjectured by J. E. Schoenfield. This also follows from the work of Higashitani and Kurimoto, except for the case $n = 21$, which they credit to a personal communication and which earlier computer searches had also settled.

1. INTRODUCTION

Let $n > k \geq 1$. For a permutation $\pi = (\pi_0, \ldots, \pi_{n-1})$ of $[n] = \{1, \ldots, n\}$, read cyclically ($\pi_{i+n} = \pi_i$), let

$$s_i = s_i(\pi) = \pi_i + \pi_{i+1} + \cdots + \pi_{i+k-1} \qquad (i \in \mathbb{Z}_n)$$

be its k-consecutive sums. They have average $k(n+1)/2$. Following Higashitani and Kurimoto [5] put

$$\mathsf{msum}(\pi, k) = \max_i s_i(\pi) - \frac{k(n+1)}{2}, \qquad \mathsf{msum}(n, k) = \min_\pi \mathsf{msum}(\pi, k).$$

Since $\max_i s_i$ is an integer at least the average, $\mathsf{msum}(n, k)$ is a non-negative integer when $k(n+1)$ is even and lies in $\frac{1}{2} + \mathbb{Z}$ when $k(n+1)$ is odd. Anstee, Ferguson and Griggs [1] studied the two-sided version $\mathsf{disc}(n, k) = \min_\pi \max_i |s_i(\pi) - k(n+1)/2| \geq \mathsf{msum}(n, k)$.

The case $k = 3$ is the *clock-face problem*: what is the least possible value $a(n)$ of the largest sum of three neighbours when $1, \ldots, n$ are placed around a circle? It goes back to Clark [3] and to a 1986 column of Martin Gardner [4], and was studied as a backtracking problem by Rolfe [7, 8]. A version with $n = 12$ was a problem of the Bundeswettbewerb Mathematik 2002 [2]: for twelve sticks of lengths $1, \ldots, 12$ in a circle, show that some three neighbours have total length at least 20. The values are OEIS sequence A066385 [6], and $a(n) = \frac{3(n+1)}{2} + \mathsf{msum}(n, 3)$. In 2019 J. E. Schoenfield observed there that $a(n) = f(n) := \lceil 3n/2 \rceil + 3$ for all $4 \leq n \leq 38$ except $n = 5, 6, 9, 15$, gave an explicit arrangement showing $a(n) \leq f(n)$ for all n, noted that $a(n) = f(n)$ is not hard to prove for even $n > 6$, and conjectured $a(n) = f(n)$ for all odd $n > 15$.

Higashitani and Kurimoto [5] determined $\mathsf{msum}(n, k)$ in many cases. Their Corollary 1.6 gives $\mathsf{msum}(n, 3) = 2$ for all odd $n \geq 7$ other than 9 and 15, which contains Schoenfield's conjecture, except that the value $\mathsf{msum}(21, 3) = 2$ is credited there to a personal communication

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of T. Yamashita [5, Remark 1.7]. For odd k and $n = mk$ with m odd they prove $\mathsf{msum}(mk, k) = 2$ for $m \geq 2k+3$ [5, Theorem 1.1], and $\mathsf{msum}(3k, k) = 1$ follows from $\mathsf{disc}(3k, k) = 1$ [1]. Since $1 \leq \mathsf{msum}(mk, k) \leq \mathsf{disc}(mk, k) \leq 2$ [1], the open cases were odd m with $5 \leq m \leq 2k+1$, except that for $k = 3$ the values $\mathsf{msum}(15, 3) = 1$ and $\mathsf{msum}(21, 3) = 2$ were known [5, Remark 1.7].

Our first result is a lower bound for odd k and all n.

Theorem 1. *Let $k \geq 3$ be odd, $n > k$ and $g = \gcd(n, k)$.*

(a) *If n is odd and $n > (k+3)g$, then $\mathsf{msum}(n, k) \geq 2$.*
(b) *If n is even and $n \neq 2k$, then $\mathsf{msum}(n, k) \geq 3/2$.*

Part (a) for $n = mk$ says $\mathsf{msum}(mk, k) \geq 2$ for odd $m \geq k+4$, improving the range $m \geq 2k+3$ of [5, Lemma 5.3]. Part (b) is included for completeness; much of it is contained in [5] (for $k = 3$ it is part of [5, Corollary 1.6(a)]), and $n = 2k$ is a genuine exception, since $\mathsf{msum}(2k, k) = \mathsf{disc}(2k, k) = \frac{1}{2}$ [1]. Our second result shows that the bound in (a) is sharp when $k \mid n$.

Theorem 2. *Let $k \geq 1$ and $m \geq 3$ be odd with $m \leq k+2$. Then there is a permutation of $[mk]$ all of whose k-consecutive sums are at most $\frac{k(mk+1)}{2} + 1$. In other words $\mathsf{msum}(mk, k) \leq 1$.*

Corollary 3. *Let $k \geq 3$ and $m \geq 3$ be odd. Then*

$$\mathsf{msum}(mk, k) = \begin{cases} 1, & m \leq k+2, \\ 2, & m \geq k+4. \end{cases}$$

For instance $\mathsf{msum}(n, 5) = 1$ for $n = 15, 25, 35$ and $\mathsf{msum}(n, 5) = 2$ for $n = 45, 55, 65, \ldots$; previously the values for $n = 25, 35, 45, 55$ were open. For $k = 3$, together with the known upper bound $a(n) \leq \lceil 3n/2 \rceil + 3$ [1, 6], we obtain the complete answer to the clock-face problem; our proof of the lower bound does not depend on any computation.

Corollary 4. *For all $n \geq 4$,*

$$a(n) = \left\lceil \frac{3n}{2} \right\rceil + 3,$$

except $a(5) = 10$, $a(6) = 11$, $a(9) = 16$ and $a(15) = 25$, which are one less. (Also $a(3) = 6$.)

The four exceptions are exactly the values of n that the inequalities of Theorem 1 leave open for $k = 3$ (namely the odd $n \leq 6\gcd(n, 3)$, and $n = 6$).

2. Slack and total variation

Throughout this section $k \geq 1$, $n > k$, and π is a permutation of $[n]$ whose k-consecutive sums satisfy $s_i \leq L$ for some integer L. Put

$$\sigma_i = L - s_i \geq 0, \qquad T = \sum_{i \in \mathbb{Z}_n} \sigma_i = nL - \frac{kn(n+1)}{2},$$

using $\sum_i s_i = k \sum_i \pi_i$. Let

$$d_i = \sigma_i - \sigma_{i+1} = s_{i+1} - s_i = \pi_{i+k} - \pi_i,$$

$$M = \sum_i \min(\sigma_i, \sigma_{i+1}), \qquad P = \#\{i : \sigma_i > 0 \text{ and } \sigma_{i+1} > 0\}.$$

Lemma 5. (i) *$d_i \neq 0$ for every i; in particular σ_i and σ_{i+1} are never both 0.*
(ii) *$\sum_i |d_i| = 2T - 2M$.*
(iii) *$M \geq P$.*

Proof. (i) As $0 < k < n$, the positions i and $i+k$ differ, so $\pi_{i+k} \neq \pi_i$. (ii) Use $|x-y| = x+y-2\min(x,y)$ and sum over i. (iii) Every pair counted by P contributes at least 1 to M, and every term of M is non-negative. □

Chains. The map $i \mapsto i+k$ splits $\mathbb{Z}_n$ into $g = \gcd(n,k)$ cycles, the *chains*, each of length $\ell = n/g \geq 2$. Along a chain $i_0, i_0+k, i_0+2k, \ldots$ the values $v_j = \pi_{i_0+jk}$ ($j \in \mathbb{Z}_\ell$) are ℓ distinct integers and $v_{j+1} - v_j = d_{i_0+jk}$. Call v_j a *peak* if $v_j > v_{j\pm1}$ and a *valley* if $v_j < v_{j\pm1}$; going around the chain, peaks and valleys alternate, so a chain with $r \geq 1$ peaks has exactly r valleys. Put

$$D = \#\{i \in \mathbb{Z}_n : d_i d_{i+k} < 0\}.$$

Since i and $i+k$ lie in the same chain, $d_i d_{i+k} < 0$ exactly when π_{i+k} is a peak or a valley of its chain. Hence $D = \sum_c 2r_c$, where $r_c \geq 1$ is the number of peaks of chain c; in particular $D \geq 2g$.

Lemma 6. $\sum_i |d_i| \geq 2n - 4g + D$.

Proof. Fix a chain with values v_j and r peaks. Write its peaks and valleys in cyclic order as $V_1, P_1, V_2, P_2, \ldots, V_r, P_r$. Its total variation is

$$\sum_j |v_{j+1} - v_j| = \sum_t (P_t - V_t) + \sum_t (P_t - V_{t+1}) = 2\sum_t P_t - 2\sum_t V_t.$$

Remove from the cyclic sequence $V_1, P_1, \ldots, V_r, P_r$ the largest value P^* (a peak) and the smallest value V^* (a valley). What remains is at most two paths (none if $r = 1$) that alternate between peaks and valleys and have an even number of terms (each path runs from a neighbour of one removed term to a neighbour of the other, and these neighbours are of opposite types). So the remaining $r-1$ peaks can be matched with the remaining $r-1$ valleys so that matched terms are adjacent in the cyclic sequence; a peak exceeds its neighbouring valleys, so each matched pair has $P - V \geq 1$. Since the ℓ values are distinct integers, $P^* - V^* \geq \ell - 1$. Therefore the total variation is at least $2(\ell-1) + 2(r-1)$. Summing over the g chains, which partition the indices i, gives $\sum_i |d_i| \geq 2g(\ell-1) + 2\sum_c (r_c - 1) = 2n - 4g + D$. □

Proposition 7. $2M + D \leq 2(T-n) + 4g$.

Proof. Combine Lemma 5(ii) and Lemma 6: $2T - 2M \geq 2n - 4g + D$. □

Lemma 8. *If k is odd, then $n - D \leq (k+1)P$.*

Proof. Let i be such that $d_i d_{i+k} > 0$; by Lemma 5(i) these are the $n - D$ indices not counted by D. Suppose that none of the $k+1$ consecutive pairs $(\sigma_{i+j}, \sigma_{i+j+1})$, $0 \leq j \leq k$, consists of two positive numbers. No such pair consists of two zeros either (Lemma 5(i)), so the numbers $\sigma_i, \sigma_{i+1}, \ldots, \sigma_{i+k+1}$ alternate between zero and positive. As k is odd, σ_{i+k} is zero exactly when σ_i is positive. If $\sigma_i > 0$, then $\sigma_{i+1} = 0$, $\sigma_{i+k} = 0$ and $\sigma_{i+k+1} > 0$, so $d_i > 0 > d_{i+k}$; if $\sigma_i = 0$, then likewise $d_i < 0 < d_{i+k}$. Both contradict $d_i d_{i+k} > 0$. Hence for each such i there is a $j \in \{i, i+1, \ldots, i+k\}$ with $\sigma_j > 0$ and $\sigma_{j+1} > 0$. A given j arises in this way from at most $k+1$ indices i (namely $i \in \{j-k, \ldots, j\}$), so $n - D \leq (k+1)P$. □

3. Proof of Theorem 1

Let $k \geq 3$ be odd and $n > k$.

(a) Let n be odd, so $k(n+1)/2$ is an integer, and suppose some π has $\mathsf{msum}(\pi, k) \leq 1$. Take $L = \frac{k(n+1)}{2} + 1$; then $T = n$, and Proposition 7 reads $2M + D \leq 4g$. As $D \geq 2g$, we get $M \leq g$ and hence $P \leq g$ by Lemma 5(iii). By Lemma 8, $D \geq n - (k+1)P$, so

$$n - (k+1)P + 2P \leq D + 2M \leq 4g, \qquad \text{i.e.} \qquad n \leq 4g + (k-1)P \leq (k+3)g.$$

So $\mathsf{msum}(n,k) \le 1$ forces $n \le (k+3)g$. Since $\mathsf{msum}(n,k)$ is an integer here, $\mathsf{msum}(n,k) \ge 2$ whenever $n > (k+3)g$.

(b) Let n be even, so $\mathsf{msum}(n,k) \in \frac{1}{2}+\mathbb{Z}$, and suppose some π has $\mathsf{msum}(\pi,k) \le \frac{1}{2}$. Take $L = \frac{k(n+1)+1}{2}$; then $T = n/2$ and Proposition 7 gives $2M + D \le 4g - n$. With $M \ge 0$ and $D \ge 2g$ this yields $n \le 2g$. Since g is odd and divides the even number n, $2g$ divides n, so $n = 2g$; and $2g = n > k$ with $g \mid k$ forces $g = k$. Hence $n = 2k$. □

Remark 9. For odd n we also have $\mathsf{msum}(n,k) \ge 1$: if $\mathsf{msum}(\pi,k) = 0$, take $L = k(n+1)/2$; then $T = 0$, all $\sigma_i = 0$ and all $d_i = 0$, contradicting Lemma 5(i).

4. The construction

Let $k, m \ge 1$. For an array $A = (A(q,c))$ with rows $q \in \mathbb{Z}_m$ and columns $c \in \{0,\dots,k-1\}$ define, for $0 \le r \le k$, the *staircase sums*

$$W_A(q,r) = \sum_{c<r} A(q+1,c) + \sum_{c\ge r} A(q,c).$$

(So $W_A(q,0)$ is the sum of row q and $W_A(q,k) = W_A(q+1,0)$.) Call A *good* if its columns are permutations of $[m]$ and $W_A(q,r) \le \frac{k(m+1)}{2} + 1$ for all q and r.

Lemma 10. *Let A be a good $m \times k$ array and β a permutation of $\{0,\dots,k-1\}$. Then $\pi_{qk+c} = m\beta(c) + A(q,c)$ $(0 \le q < m,\ 0 \le c < k)$ defines a permutation of $[mk]$ all of whose k-consecutive sums are at most $\frac{k(mk+1)}{2} + 1$.*

Proof. Column c of A takes each value of $[m]$ once, so π takes each value of the block $\{m\beta(c)+1,\dots,m\beta(c)+m\}$ once, and these blocks partition $[mk]$. The k consecutive positions starting at $qk+r$ $(0 \le r < k)$ are $qk+c$ for $c \ge r$ and $(q+1)k+c$ for $c < r$ (positions modulo mk, rows modulo m), one in each column. Their sum is $m\sum_c \beta(c) + W_A(q,r) = \frac{mk(k-1)}{2} + W_A(q,r) \le \frac{mk(k-1)}{2} + \frac{k(m+1)}{2} + 1 = \frac{k(mk+1)}{2} + 1$. □

Lemma 11 (base case)**.** *Let $m = 2h+1 \ge 3$ and $k = m-2$. Define $C : \mathbb{Z}_m \to [m]$ by $C(i) = m-i$ for $0 \le i \le h$ and $C(i) = i-h$ for $h < i \le 2h$, so that $C = (m, m-1, \dots, h+1, 1, 2, \dots, h)$. Then $A(q,c) = C(q+hc)$ is a good $m \times k$ array.*

Proof. C is a bijection onto $[m]$, so every column is a permutation of $[m]$. Since $2h = m-1 \equiv -1 \pmod m$ we have $1 \equiv -2h$, hence $A(q+1,c) = C(q+h(c-2))$. Therefore the entries of $W_A(q,r)$ are the values $C(q+hj)$ for

$$j \in \{-2,-1,\dots,r-3\} \cup \{r, r+1, \dots, k-1\} = \{-2,\dots,k-1\} \setminus \{r-2, r-1\}.$$

The set $\{-2,\dots,k-1\}$ consists of $k+2 = m$ consecutive integers and $\gcd(h,m) = 1$, so the residues $q+hj$, $-2 \le j \le k-1$, run over all of $\mathbb{Z}_m$. With $y = q + h(r-2)$ we obtain

$$W_A(q,r) = \sum_{x\in\mathbb{Z}_m} C(x) - C(y) - C(y+h) = \frac{m(m+1)}{2} - \big(C(y) + C(y+h)\big).$$

Now $C(y) + C(y+h) \ge m$ for every y: for $1 \le y \le h$ it equals $(m-y) + y = m$; for $y = 0$ it equals $m + (h+1)$; and for $h < y \le 2h$ we have $y + h \equiv y - h - 1 \in \{0,\dots,h-1\}$, so it equals $(y-h) + (m-y+h+1) = m+1$. Hence $W_A(q,r) \le \frac{m(m+1)}{2} - m = \frac{(m-2)(m+1)}{2} + 1 = \frac{k(m+1)}{2} + 1$. □

Lemma 12 (adding two columns)**.** *Let A be a good $m \times k$ array and $0 \le j < k$. Insert immediately before column j two new columns, first a copy U of column j and then its complement $\bar{U} = m + 1 - U$. The resulting $m \times (k+2)$ array A' is good.*

Proof. The new columns are permutations of $[m]$. In A' the columns are, in order, the old columns $0,\dots,j-1$, then $U,\bar U$, then the old columns $j,\dots,k-1$. Consider a staircase sum $W_{A'}(q,r')$, $0\le r'\le k+2$. If $r'\le j$, both new columns are taken from row q; if $r'\ge j+2$, both are taken from row $q+1$. In these cases they contribute $U+\bar U=m+1$, and the old columns contribute $W_A(q,r')$ or $W_A(q,r'-2)$ respectively, so $W_{A'}(q,r')\le \frac{k(m+1)}{2}+1+(m+1)=\frac{(k+2)(m+1)}{2}+1$. If $r'=j+1$, then U is taken from row $q+1$ and $\bar U$ from row q, while the old columns contribute $W_A(q,j)$; thus

$$W_{A'}(q,j+1)=W_A(q,j)+A(q+1,j)+\big(m+1-A(q,j)\big)=W_A(q,j+1)+(m+1),$$

and the same bound holds. □

Proof of Theorem 2. Start from the good $m\times(m-2)$ array of Lemma 11 and apply Lemma 12 $\frac{k-m+2}{2}$ times to obtain a good $m\times k$ array; then apply Lemma 10. □

For example, for $k=m=5$, inserting the pair before column $j=1$ of the base array for $m=5$ gives the columns 54312, 31254, 35412, 31254, 25431 (each listing rows 0 to 4), and $\beta(c)=4-c$ gives the arrangement

$$25,18,13,8,2,\ 24,16,15,6,5,\ 23,17,14,7,4,\ 21,20,11,10,3,\ 22,19,12,9,1$$

of [25], whose 5-consecutive sums are 63, 65 or 66, while the average is 65. In general the staircase sums of the arrays in Theorem 2 take only the three values $\frac{k(m+1)}{2}+1$, $\frac{k(m+1)}{2}$ and $\frac{k(m+1)}{2}-\frac{m-1}{2}$.

5. Proofs of the corollaries

Proof of Corollary 3. By Remark 9, $\mathsf{msum}(mk,k)\ge 1$, and by Theorem 2 equality holds for $m\le k+2$. For $m\ge k+4$ we have $g=\gcd(mk,k)=k$ and $mk>(k+3)k$, so Theorem 1(a) gives $\mathsf{msum}(mk,k)\ge 2$, while $\mathsf{msum}(mk,k)\le \mathsf{disc}(mk,k)\le 2$ for odd k by [1] (see [5, Theorem 2.4(iv)]). □

Proof of Corollary 4. Here $a(n)=\frac{3(n+1)}{2}+\mathsf{msum}(n,3)$, and $f(n)=\lceil 3n/2\rceil+3$ equals $\frac{3(n+1)}{2}+2$ for odd n and $\frac{3(n+1)}{2}+\frac{3}{2}$ for even n.

Upper bound. For $n\ge 6$, $\mathsf{msum}(n,3)\le \mathsf{disc}(n,3)\le 2$ [1] (see [5, Theorem 2.4(iii)]); for even n this means $\mathsf{msum}(n,3)\le\frac{3}{2}$ because $\mathsf{msum}(n,3)\in\frac{1}{2}+\mathbb{Z}$. So $a(n)\le f(n)$ for $n\ge 6$. Schoenfield's explicit arrangements [6] give the same bound for all $n\ge 3$. For $n=4$, every three neighbours omit one number, so $a(4)=10-1=9=f(4)$.

Lower bound. Let $n\ge 5$ and $g=\gcd(n,3)$. If n is odd and $3\nmid n$, Theorem 1(a) gives $\mathsf{msum}(n,3)\ge 2$ for $n>6$. If $n=3m$ is odd, Corollary 3 gives $\mathsf{msum}(n,3)=2$ for $m\ge 7$, that is for $n\ge 21$. If n is even, Theorem 1(b) gives $\mathsf{msum}(n,3)\ge\frac{3}{2}$ unless $n=6$. Hence $a(n)=f(n)$ for all $n\ge 4$ except possibly $n=5,6,9,15$.

The exceptions. For odd n, $a(n)\ge\frac{3(n+1)}{2}+1=f(n)-1$ by Remark 9, and for $n=6$, $a(6)\ge\lceil 21/2\rceil=11=f(6)-1$. The arrangements

$$(5,1,4,3,2),\qquad (6,1,4,5,2,3),\qquad (9,4,3,8,5,2,7,6,1),$$
$$(15,8,2,14,6,5,13,7,4,11,10,3,12,9,1)$$

have largest sums of three neighbours 10, 11, 16 and 25, respectively; the last two come from Theorem 2 with $(k,m)=(3,3)$ and $(3,5)$. □

6. Computations

The arguments above need no computer. As a check we verified the following by computer; the programs are short C and Python codes.

(1) Lemma 5, Lemma 6, Proposition 7, Lemma 8 and the final inequality $n \le 2(T-n) + 4g + (k-1)P$ hold for every circular arrangement of $[n]$ with $4 \le n \le 11$ ($k = 3$) and $6 \le n \le 11$ ($k = 5$), taking for L the largest k-consecutive sum and the next two integers (about 2.4×10^7 cases in all). A second, independently written program confirmed them for all $k \le 9$ with $n \le 10$ and for $n = 11, 12$ with $k = 3, 5, 7, 9$ (Lemma 8 and the final inequality for odd k only).

(2) The arrays of Theorem 2, with several insertion positions and random block orders β, give arrangements with largest k-consecutive sum exactly $\frac{k(mk+1)}{2} + 1$ for all odd $3 \le k \le 61$ and odd $3 \le m \le k+2$; the second program also tried every sequence of insertion positions for $k \le 11$ and every block order β for $k \le 7$.

(3) An exhaustive branch-and-bound search, which prunes with Lemmas 5(ii) and 6 and was checked against plain enumeration, reproduces all published values $a(3), \ldots, a(38)$ and gives $a(39) = 62$, $a(40) = 63$, $a(41) = 65$, $a(42) = 66$, $a(43) = 68$ and $a(44) = 69$, in agreement with Corollary 4. For $k = 5$ it finds arrangements with $\mathsf{msum}(\pi, 5) = 1$ for $n = 15, 25, 35$, in agreement with Corollary 3.

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